

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EC)

December 2011

EC 305 Microcontroller & Application

Date: 16.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Answer the following questions with respect to 8051 microcontroller (μC): [07]

1. Determine maximum delay that can be provided by timer1 in mode-1 with a crystal of 12 MHz.
2. What is the function of ALE signal?
3. What is the function of PSEN pin of 8051?
4. Why pull-up resistors are necessary with Port 0?
5. What is the function of GATE bit in TMOD register?
6. Give difference between PC and DPTR pointers.
7. List the addressing modes of 8051 assembly programming.

Q - 2 (a) Find the TMOD register value for timer 0 and timer 1, mode 2, software start/stop, with the clock coming from the 8051's crystal. [04]

(b) What are the interrupts available in 8051 microcontroller? Explain Interrupt Enable and Interrupt Priority registers format. [06]

(c) Explain different types of flags and significance of each flag status in 8051. [04]

OR

Q - 2 (a) Upon activation of an interrupt, what are the steps that the μC needs to follow? [06]

(b) Give differences between 'Level-triggered interrupt' and 'Edge-triggered interrupt'. [04]

(c) To monitor the timer flag (TF) the μC is tied down in polling and cannot do anything else. Can we implement the interrupt to watch the TF bit? Briefly explain. [04]

Q - 3 (a) Interface single 64 K ROM with 8051. Assume that from address 0000H to 7FFFH program codes are stored in ROM and from address B000H to FFFFH data are stored in ROM. To access program code and data which instructions are used by 8051? [07]

(b) In 8051 serial communication, how serial flags are useful to read / write data byte in SBUF register successfully. [07]

OR

Q - 3 (a) Draw & explain the block diagram of interfacing 8031 μ C with external RAM and ROM. [07]

(b) What are the registers used for serial communication in 8051. Discuss the serial communication modes 2 and 3. [07]

SECTION – II*

Q - 4 (a) Describe the function of following instructions of 8051. Which flags are affected after execution of the instruction? [07]

- | | |
|---------|----------|
| 1. DA A | 5. RETI |
| 2. XCHD | 6. ACALL |
| 3. RR | 7. LCALL |
| 4. LJMP | |

Q - 5 (a) Write a program to generate 2 KHz square wave on pin P1.7 of port 1 of 8051 μ C using Interrupt Service Routine. [08]

(b) Calculate the Timer 0 reload value needed to achieve time delay of 20 ms. Oscillator frequency is 11.0592 MHz. [06]

OR

Q - 5 (a) Write a 'C'- program for 8051 μ C to toggle pin 7 on Port 1 with a time delay of 20 ms. Crystal frequency is 11.0592 MHz. [08]

(b) Find the TH1 value (in both decimal and hex) to set the baud rate of 9600 and 4800 if SMOD=1. Crystal frequency is 11.0592 MHz. [06]

Q - 6 (a) With a neat circuit diagram explain how a 4 X 4 keypad is interfaced with 8051 μ C? [10]
Write an assembly level program for keypad scanning. Consider a problem of 'Switch Bounce' because of the spring action of the switch.

(b) Find the Iout for the following inputs. Assume Iref = 2 mA for DAC080. [04]
1) 99H and 2) C8H

OR

Q - 6 (a) Assume that the INT1 pin is connected to a switch that is normally high. Whenever it goes low, it should turn on an LED. The LED is connected to P1.3 and is normally off. When it is turned on it should stay on for a fraction of a second. As long as the switch is pressed low, the LED should stay on. Write an assembly program and draw a circuit diagram. [10]

(b) Write an advantage and a disadvantage of using an IC chip for keyboard scanning and decoding instead of using a μ C. [04]